

Student Resource Guide

Generative AI for Beginners

Section 5: Audio and Video Generation | Lecture 5.4: Deepfakes and Face Synthesis

Learning Objectives

- Explain the technical mechanisms behind face-swapping and face reenactment deepfakes
- Identify the legitimate and harmful applications of face synthesis technology
- Describe deepfake detection approaches and their current limitations
- Evaluate the ethical, legal, and organisational responsibilities of practitioners deploying face synthesis
- Develop a framework for responsible engagement with face synthesis technology in professional contexts

Introduction

Deepfake technology represents one of the most powerful illustrations of the dual-use nature of generative AI. The same algorithms that enable legitimate applications—personalised digital avatars, entertainment content, accessibility tools for people who have lost their ability to speak—also enable convincing impersonation, non-consensual imagery, and political disinformation. For AI practitioners, understanding both the technical mechanisms and the ethical framework for responsible engagement with this technology is not optional—it is a professional responsibility.

The technical sophistication required to create convincing deepfakes has decreased dramatically over the past decade. What required a research team with specialised hardware in 2017 can now be produced using consumer applications, open-source tools, and commodity GPUs. This democratisation of capability means that practitioners in any industry may encounter deepfake-related challenges—both in deploying legitimate face synthesis tools and in detecting and responding to malicious deepfakes targeting their organisation or clients.

This guide extends the lecture's technical content with a focus on practitioner responsibilities: detection capability, governance frameworks, consent requirements, and the emerging legal landscape. Face synthesis is one of the highest-risk AI application categories, and approaching it with appropriate rigour is essential to responsible practice.

Key Concepts

Face Swapping Architecture

Definition	A deepfake technique using shared-encoder autoencoders: two autoencoders share a common encoder that learns pose, expression, and lighting representations, but have separate decoders trained on different identities. Face swapping occurs by encoding the source face (capturing its pose/expression) and decoding through the target identity's decoder (applying the target's facial features).
Business Context	The shared-encoder architecture is why face swapping requires relatively little training data per identity (hours of video rather than thousands of photographs)—the encoder's general face understanding is shared; only the decoder personalises to each identity. This accessibility is both a feature (legitimate use cases) and a risk (low barrier to misuse).
AI Practitioner relevance	Practitioners building digital avatar systems or virtual presenter tools should understand that the same architectural components used in legitimate applications are used in malicious deepfakes. This creates reputational and legal risk if tools are deployed without adequate consent frameworks and misuse prevention controls.
Real-World Example	DeepFaceLab, an open-source face-swapping tool, uses the shared-encoder architecture. A film production company legitimately used it to de-age an actor's face in historical flashback scenes, blending AI-generated face swaps with traditional makeup—a production technique now common in major studio productions.

Face Reenactment

Definition	A technique that animates a target person's face based on the facial movements of a different 'driving' video or 3D model, without replacing the target's identity. The target face maintains its appearance while adopting the pose, expression, and
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	movements of the driving source.
Business Context	Face reenactment enables cost-effective video localisation (dubbing a spokesperson's face to match dubbed audio lip movements), animation of historical photographs, and accessibility applications (reanimating a person's still photograph into a speaking video using audio input).
AI Practitioner relevance	Practitioners evaluating face reenactment for video localisation should assess lip-sync quality, expression naturalness, and edge-case handling (glasses, beards, extreme expressions). Commercial tools (D-ID, HeyGen, Synthesia) offer reenactment as a managed service with consent verification requirements built into their terms of service.
Real-World Example	HeyGen's video translation feature uses face reenactment to reanimate a speaker's face to match dubbed audio in a new language—preserving the original speaker's identity and emotional expression while synchronising lip movements to the translated speech. This is used extensively for multilingual corporate video production.
Key Risks	Face reenactment without consent is used to create false statements attributed to real people. In political contexts, reenactment of political figures saying things they did not say has been used in disinformation campaigns. Consent and content policy enforcement are non-negotiable for any commercial deployment.

GAN-Based Face Synthesis for Photorealistic Avatars

Definition	Using trained GAN architectures (StyleGAN, First Order Motion Model) to generate photorealistic human face images and animate them. Combines a high-quality face generator (StyleGAN3) with a motion transfer model to produce fully synthetic, non-existent human avatars that can be animated to speak, express emotions, and interact.
Business Context	Fully synthetic AI avatars—faces of people who do not exist—avoid consent issues entirely and are used commercially for

	customer service representatives, training video presenters, news readers, and virtual brand ambassadors. Companies like Synthesia and D-ID offer this capability as a service.
AI Practitioner relevance	When using synthetic avatars, practitioners should disclose to audiences that they are interacting with an AI-generated representation, not a real person. Several emerging regulations (EU AI Act) and platform policies (YouTube, Meta) require disclosure of synthetic media—build disclosure into the content design from the start.
Real-World Example	A global financial services firm produces compliance training videos using a Synthesia AI avatar presenter. The avatar is disclosed as AI in the video's opening frame. 30 training courses in 12 languages are produced without any human presenter filming time—reducing production cost by 80% vs. live-action filming.

Deepfake Detection

Definition	Technical approaches to identifying synthetic or manipulated media, including: biological signal analysis (natural blinking patterns, pulse signals in skin), artifact detection (GAN-specific artifacts, boundary inconsistencies, unnatural eye reflections), provenance tracking (digital watermarks, C2PA content credentials), and classifier-based detection models trained on deepfake datasets.
Business Context	Deepfake detection is a necessary defensive capability for organisations that: operate voice or face authentication systems, receive video evidence in legal or compliance contexts, publish video content requiring source authentication, or are potential targets of executive impersonation fraud.
AI Practitioner relevance	Practitioners responsible for content integrity or security should be aware that deepfake detection is an arms race—detection models are trained on current deepfake techniques and lag behind new generation methods. No detection tool

	achieves 100% accuracy. A defence-in-depth approach combining technical detection, provenance standards, and human review is more reliable than any single detection tool.
Real-World Example	Microsoft Video Authenticator and Intel's FakeCatcher provide deepfake detection as APIs. FakeCatcher uses blood flow signals (rPPG) visible in facial video to detect synthetic media—real faces show subtle skin colour changes as blood flows; synthetic faces typically do not reproduce this signal. Detection accuracy is approximately 96% on benchmark datasets but varies in real-world conditions.

Real-World Applications

Face synthesis has a legitimate commercial ecosystem and a parallel misuse ecosystem. Both are relevant to practitioner awareness.

Synthetic AI Presenter Production: Corporate training, e-learning, and corporate communications teams use synthetic avatar services (Synthesia, D-ID) to produce video content at scale. A presenter avatar is created once (from a consented actor or a fully synthetic AI face) and then used to generate thousands of videos by supplying new scripts. Production cost is 80–95% lower than live filming.

Film and Video Post-Production: Professional film productions use face swapping and de-aging technology for legitimate creative purposes: replacing stunt performers' faces with actors, de-aging actors for flashback scenes, and recreating historical figures. These techniques are subject to professional ethics frameworks, union agreements, and explicit consent contracts.

Accessibility: Facial Animation from Audio: For individuals who have lost mobility but retain vocal ability, or who have lost vocal ability but retained facial mobility, face reenactment technology enables new forms of communication. Applications animate facial expressions from audio input or reanimate photographs for video communication.

Deepfake Fraud and Disinformation (Misuse Context): This is documented for awareness, not implementation. Deepfakes are used in: CEO fraud (phone/video calls impersonating executives to authorise transfers), non-consensual intimate imagery (the most common misuse), election disinformation (fake political statements), and identity theft. Practitioners must understand misuse patterns to implement effective detection and prevention.

Best Practices

Obtain written, scoped consent before using any real person's likeness. Consent must specify: which platforms the content will appear on, for how long, for what purpose. Generic consent for 'AI use' is insufficient. Obtain separate consent for training data use and for each output use case.

Disclose synthetic media to audiences consistently. Label AI-generated or AI-modified video content as such in metadata (C2PA standards), in on-screen text, and in platform disclosure fields. Build disclosure into the template, not as a manual afterthought.

Implement deepfake detection in authentication systems. If your organisation uses video or voice for identity verification, integrate deepfake detection as a parallel layer. Do not rely on a single detection technology; layer multiple signals.

Maintain audit logs of all synthetic media produced. Record what was generated, when, using which model, by whom, and for what purpose. Retention periods should align with relevant statutes of limitations—typically 3–7 years for commercial content.

Refuse requests to create synthetic media of public figures without unambiguous satirical framing. Creating realistic synthetic video of politicians, executives, or public figures making statements they did not make—even for 'illustrative' or 'educational' purposes—creates significant legal and reputational risk.

Common Mistakes to Avoid

Mistake: Assuming synthetic avatar tools manage consent and disclosure compliance on behalf of the user.

Correct approach: Commercial synthetic media tools provide the technology; compliance with consent, disclosure, and content regulations is the deployer's responsibility. Review your jurisdiction's requirements independently and implement appropriate controls.

Mistake: Treating deepfake detection as a binary solved problem.

Correct approach: Deepfake detection is probabilistic and an ongoing arms race. No tool achieves 100% detection accuracy across all current techniques. Implement detection as one layer in a defence-in-depth strategy; never rely on it as the sole control.

Mistake: Believing that using a synthetic face (rather than a real person) eliminates all ethical concerns.

Correct approach: Fully synthetic faces avoid consent issues related to specific individuals, but other ethical concerns remain: demographic representation and bias in

synthetic faces, disclosure obligations to audiences, and the broader social impact of normalising synthetic human presence in media.

Practitioner Tool: Face Synthesis Deployment Governance Framework

Type: Governance Checklist

Complete this checklist before deploying any face synthesis, face reenactment, or AI avatar system in a production context.

- Confirm whether real human likenesses or fully synthetic faces are used—apply consent requirements for real faces
- Obtain written, scoped consent from all individuals whose likeness will be synthesised
- Define content scope: what the synthesised content may show, say, or depict
- Implement C2PA content credentials or equivalent metadata watermarking
- Include on-screen or platform disclosure labels on all synthetic content
- Establish content moderation review for all outputs before publication
- Implement audit logging (what was generated, by whom, when, for what purpose)
- Define and document prohibited content types (false statements, non-consensual intimate content)
- Verify jurisdictional compliance (EU AI Act, deepfake-specific state laws in USA, etc.)
- Train content production team on consent requirements and prohibited use cases
- Establish incident response procedure for deepfake misuse allegations
- Review and update governance framework quarterly as regulations evolve

Note: This checklist should be completed by a cross-functional team including legal, compliance, and product management—not by the technical implementer alone.

Knowledge Check

Question 1: A company uses a face-swapping tool to create a training video where a celebrity's face is replaced with a synthetic face to illustrate a scenario. No consent was obtained from the celebrity. What is the primary governance failure?

- A. The company failed to watermark the output with deepfake detection markers
- **B. The company failed to obtain consent from the individual whose likeness was used for face swapping**
- C. The company should have used a GAN-based tool rather than an autoencoder
- D. Face-swapping tools are prohibited for corporate training use

Correct Answer: B. The company failed to obtain consent from the individual whose likeness was used for face swapping

Explanation: Using a real person's likeness in face synthesis without consent violates ethical standards and potentially privacy law. Consent is required regardless of the intended use case or whether the face is ultimately visible in the output.

Question 2: An organisation's security team identifies a suspicious video call where an executive appears to authorise a large financial transfer. Deepfake detection software flags the video as 75% likely synthetic. What is the appropriate response?

- A. Proceed with the transfer since the detection confidence is only 75%
- **B. Block the transfer and verify the executive's identity through an independent, out-of-band communication channel**
- C. Ignore the flag—deepfake detection tools have too many false positives to act on
- D. Request a higher-resolution video call to improve detection accuracy

Correct Answer: B. Block the transfer and verify the executive's identity through an independent, out-of-band communication channel

Explanation: A 75% deepfake probability flag on a high-value financial transaction is sufficient to pause and verify through independent means. The correct response is to verify identity out-of-band (phone call to a known number, in-person confirmation) before proceeding—never to proceed with a flagged high-value action.

Reflection Questions

Take time to consider the following questions before the next session:

1. Face synthesis technology enables both the preservation of a person's likeness after death and the malicious impersonation of living people. How should legal and ethical frameworks distinguish between these uses?
2. Detection technology consistently lags behind generation technology in the deepfake arms race. What non-technical mechanisms (provenance standards, platform policy, regulation) provide complementary protection?
3. Non-consensual deepfake intimate imagery causes documented psychological harm to victims. What systemic responses—technical, legal, and platform-level—are needed, and what role can practitioners play?
4. If synthetic presenter avatars become ubiquitous in corporate communications, what are the implications for human trust, employee relations, and organisational authenticity?
5. C2PA content credentials (a standard for provenance metadata) require all parties in the content chain to implement the standard to be effective. What does this tell you about the limits of technical standards as a governance mechanism?

Key Takeaways

- Face swapping uses shared-encoder autoencoders: a common encoder learns pose/expression; separate decoders reconstruct each identity from these representations.
- Face reenactment animates a target face using a driving video's motion without replacing the target's identity—used legitimately for video localisation and problematically for false statement creation.
- Fully synthetic AI avatars (non-existent faces) avoid individual consent requirements but retain disclosure obligations and demographic representation considerations.
- Deepfake detection is probabilistic and lags behind generation technology—use it as one layer in a defence-in-depth strategy, not as a binary filter.
- Every face synthesis deployment requires: written consent (for real faces), content scope definition, disclosure labelling, audit logging, and prohibited content policies.
- The deepfake misuse landscape (CEO fraud, disinformation, NCII) requires practitioners to treat face synthesis as a high-risk AI capability requiring elevated governance regardless of the legitimate use case.

Next Actions

Complete at least two of the following before the next lecture:

6. Review your organisation's current authentication procedures for identifying whether executive video/voice communications could be verified independently if deepfake-suspected.
7. Investigate one commercial synthetic avatar platform (e.g., Synthesia) and document their consent requirements, disclosure policies, and content moderation approach.
8. Research C2PA (Coalition for Content Provenance and Authenticity) content credentials standard—understand how it works and which platforms currently support it.
9. Locate your jurisdiction's current legal status on deepfakes (EU AI Act provisions, USA state laws, or equivalent) and document the key compliance requirements for your organisation.